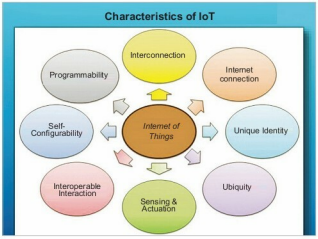

Figure 1: Characteristics of IoT (Image source: *googleimages.com*)

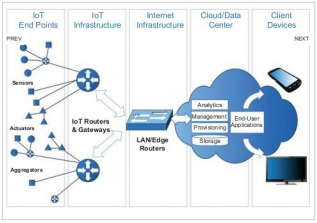

Figure 2: IoT node types and the data paths (Image source: *googleimages.com*)

The challenges of IoT

Though IoT usage is growing in different fields, there are a few challenges that require our attention and action so that it can be efficiently leveraged in areas where it is not yet being used. Let's have a look at some of these challenges.

Data storage and analytics: One of the challenges for developers of IoT applications is to clean, process and then interpret the large amounts of data gathered by different sensors. The proposed solution for this is to use wireless sensor networks. These networks share all the data that is gathered by sensor nodes, which is then sent to a distributed system for analytics. Another challenge is the storage of such large volumes of data.

Platform fragmentation: IoT also suffers from platform fragmentation (the inability to support a large number of platforms) and lack of common technical standards. Currently, a wide variety of IoT devices (in terms of both hardware and the differences in the software running on them) makes the task of developing different applications that work consistently across different inconsistent technical systems, difficult. Customers may be a bit hesitant to bet their IoT future on proprietary software or different hardware devices that use proprietary protocols in the fear that these may become obsolete or be difficult to customise.

Privacy, autonomy and control: Although IoT has immense potential to empower citizens by making governments transparent and by broadening information access, there are also serious threats to a citizen's privacy and the scope for political manipulation by the state. Such concerns have led many to conclude that different Big Data infrastructures like the kind required for the Internet of Things and for data mining are incompatible with privacy.

Security: There have been many concerns raised that IoT is being developed rapidly without much thought being given to the profound security challenges associated with it and the different regulatory changes that might be necessary. When we talk of security concerns related to IoT, we refer to securing servers and workstations. The common measures like firewalling or security updates are unsuitable for much smaller IoT devices.

Design: The design and management of IoT must be sustainable and secure. The design of IoT devices must factor in uncertain futures with respect to their management, without risking physical failure. We cannot consider IoT devices to be successful without giving due consideration to the interface's usability as well as the technology. The interfaces need to be not just user friendly but also better integrated.

Complexity and unclear value propositions: According to the feedback of several users, IoT solutions are either too complex or lack a clear use case for different end users. Experts also say that the IoT industry is currently heavily focused on gadgets, and is not making those gadgets relevant to particular business verticals. There are many who are just not able to pinpoint what value IoT offers them.

Traditional governance structures: There is a clash between IoT and the companies' traditional governance structures, as IoT still presents both uncertainties and the lack of historical precedence. Definite processes are needed to capture the IoT opportunity. This will help to improve the organisational design processes and to test the new innovation management practices.

Role of open source in handling the challenges and other aspects of IoT

The impact of IoT will be felt across a wide range of industries and applications, including agriculture, medical, manufacturing, electronics, consumer, transportation and energy. Just like the Internet, the emerging IoT will also rely upon and promote the adoption of different open source technologies and standards, as without the adoption of open standards and protocols, our devices may not be able to communicate with one another. While many of the IoT devices ultimately connect back to the Internet, the different methods they use to communicate with one another and with the local control hubs are